

ALLIANCE UNIVERSITY

DBMS ASSIGNMENT 2

Entity Relationship (ER) Diagram and Relational Schema

Details	Information
Application / Domain	College Management System
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Course / Section	BCA SEM 3 2025
Based On	First DBMS Assignment

Objective: Design a complete Entity Relationship (ER) Diagram for the same College Management System selected in the First DBMS Assignment and convert it into a Relational Schema.

1. List of Entities and Attributes

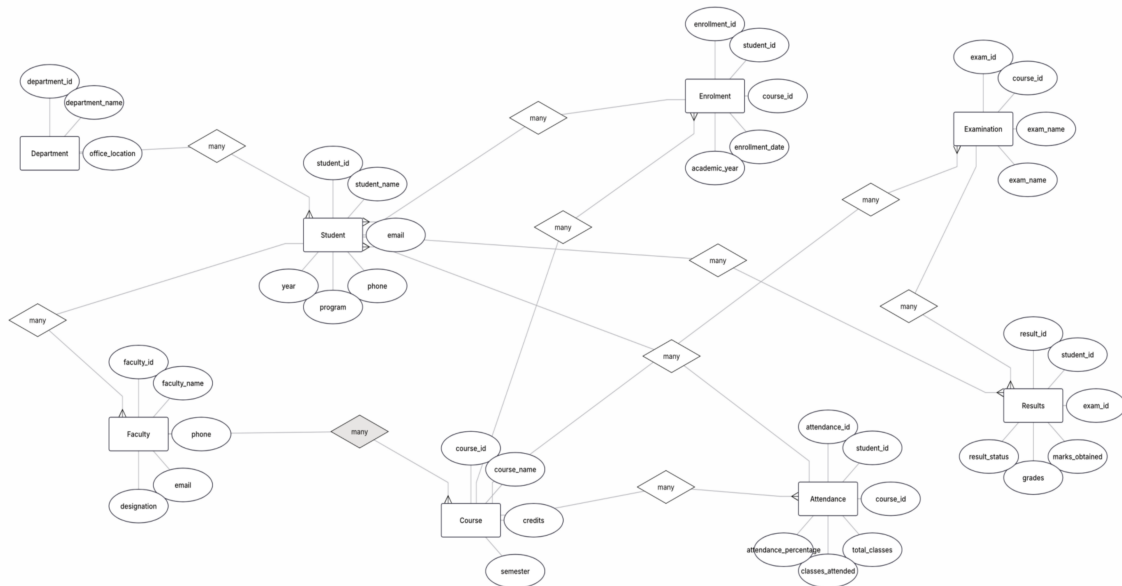
This assignment uses the same eight core tables identified in the First DBMS Assignment: Departments, Students, Faculty, Courses, Enrollments, Attendance, Examinations, and Results.

Entity	Attributes	Primary Key
Departments	department_id, department_name, hod_name, office_email	department_id
Students	student_id, student_name, date_of_birth, gender, email, phone, department_id, admission_year	student_id
Faculty	faculty_id, faculty_name, email, phone, designation, department_id	faculty_id
Courses	course_id, course_name, credits, semester, department_id, faculty_id	course_id
Enrollments	enrollment_id, student_id, course_id, enrollment_date, academic_year	enrollment_id
Attendance	attendance_id, student_id, course_id, total_classes, classes_attended, attendance_percentage	attendance_id
Examinations	exam_id, course_id, exam_name, exam_date, max_marks	exam_id
Results	result_id, student_id, exam_id, marks_obtained, grade, result_status	result_id

2. Relationships and Cardinalities

Relationship	Cardinality	Explanation
Departments → Students	1 : M	One department can have many students, while each student belongs to one department.
Departments → Faculty	1 : M	One department can have several faculty members, while each faculty member belongs to one department.
Departments → Courses	1 : M	A department can offer multiple courses, while each course belongs to one department.
Faculty → Courses	1 : M	One faculty member can teach multiple courses, while each course is assigned to a faculty member.
Students → Enrollments	1 : M	One student can have several enrollment records.
Courses → Enrollments	1 : M	One course can have many enrolled students.
Students → Attendance	1 : M	A student can have attendance records for multiple courses.
Courses → Attendance	1 : M	A course can have attendance records for many students.
Courses → Examinations	1 : M	A course can have multiple examinations.
Students → Results	1 : M	A student can have multiple examination results.
Examinations → Results	1 : M	One examination can produce results for many students.
Students ↔ Courses	M : N	Students and courses have a many-to-many relationship, resolved through the Enrollments table.

3. ER Diagram



4. Convert into Relational Schema

The following relational schema preserves the eight tables and foreign-key connections from the First DBMS Assignment.

Table	Relational Schema
DEPARTMENTS	DEPARTMENTS(department_id PK, department_name, hod_name, office_email UNIQUE)
STUDENTS	STUDENTS(student_id PK, student_name, date_of_birth, gender, email UNIQUE, phone, department_id FK, admission_year)
FACULTY	FACULTY(faculty_id PK, faculty_name, email UNIQUE, phone, designation, department_id FK)
COURSES	COURSES(course_id PK, course_name, credits, semester, department_id FK, faculty_id FK)
ENROLLMENTS	ENROLLMENTS(enrollment_id PK, student_id FK, course_id FK, enrollment_date, academic_year)
ATTENDANCE	ATTENDANCE(attendance_id PK, student_id FK, course_id FK, total_classes, classes_attended, attendance_percentage)
EXAMINATIONS	EXAMINATIONS(exam_id PK, course_id FK, exam_name, exam_date, max_marks)
RESULTS	RESULTS(result_id PK, student_id FK, exam_id FK, marks_obtained, grade, result_status)

Primary Key and Foreign Key Identification

- DEPARTMENTS.department_id — Primary Key.
- STUDENTS.student_id — Primary Key; department_id is a Foreign Key referencing DEPARTMENTS.department_id.
- FACULTY.faculty_id — Primary Key; department_id is a Foreign Key referencing DEPARTMENTS.department_id.
- COURSES.course_id — Primary Key; department_id references DEPARTMENTS.department_id and faculty_id references FACULTY.faculty_id.
- ENROLLMENTS.enrollment_id — Primary Key; student_id references STUDENTS.student_id and course_id references COURSES.course_id.
- ATTENDANCE.attendance_id — Primary Key; student_id references STUDENTS.student_id and course_id references COURSES.course_id.
- EXAMINATIONS.exam_id — Primary Key; course_id references COURSES.course_id.
- RESULTS.result_id — Primary Key; student_id references STUDENTS.student_id and exam_id references EXAMINATIONS.exam_id.

5. Relationship Explanation

Departments provide the organizational structure for students, faculty and courses. Each of these relationships is one-to-many. Faculty members teach courses, giving a one-to-many Faculty–Courses relationship in the simplified model. Students and Courses are many-to-many because a student can study multiple courses and a course can have multiple students; this is represented through ENROLLMENTS. Attendance records connect students and courses, while Examinations belong to courses and Results connect students with examinations. Thus, the foreign keys maintain the connections among all eight tables.

6. Conclusion

The ER model and relational schema provide a structured representation of the same College Management System defined in the First DBMS Assignment. The design contains the eight original tables—Departments, Students, Faculty, Courses, Enrollments, Attendance, Examinations and Results—and preserves their primary keys, foreign keys and stated cardinalities.